

BracU is going to introduce two types of courses for the Spring 23 semester, online & offline. Design a system that will determine the eligibility of a student to do an online or an offline course based on the following criteria:

Inputs

- 1) **U**: Current Location (U = 1: student is at university, U = 0: at home)
- 2) **L**: Laptop (L = 1, student has a laptop, L = 0 means otherwise)
- 3) **H**: Headphones (H = 1: has headphones, H = 0 means otherwise)
- 4) **V**: Vaccine (V = 1: student is vaccinated, V = 0 means otherwise)

Outputs

A: Offline Course

B: Online Course

The outputs will depend on the following conditions:

- Offline courses can be taken if the student is present in the university and either s/he doesn't have a laptop or headphones.
- Students must be vaccinated to attend an offline course.
- For an online course, if at home the student must have both a laptop and headphones.
- Students can also attend an online class from university if s/he is vaccinated, or s/he has a laptop with headphones
- The system can suggest both online & offline class.

a) Devise a truth table for the system above which will suggest type of course based on the given inputs. In the truth table, maintain **ULHV** sequence, U is the MSB, V is the LSB.

b) Use K-Maps to derive a simplified equations for both of the outputs.

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- **Offline** courses can be taken if the student is present in the university and either s/he doesn't have a laptop or headphones.
- Students must be vaccinated to attend an **offline** course.

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- **Offline** courses can be taken if the student is in the university and either s/he doesn't have a laptop or headphones.

$$\mathbf{U = 1 \text{ AND } (L = 0 \text{ OR } H = 0)}$$

- Students must be vaccinated to attend an **offline** course. **AND V = 1**

A = 1 (Offline = 1) When:

U = 1 AND (**L = 0** OR **H = 0**) AND **V = 1**

#	U	L	H	V	A	B
0	0	0	0	0	00	
1	0	0	0	1	0	
2	0	0	1	0	00	
3	0	0	1	1	0	
4	0	1	0	0	00	
5	0	1	0	1	0	
6	0	1	1	0	00	
7	0	1	1	1	0	
8	1	0	0	0	0	
9	1	0	0	1	1	
10	1	0	1	0	0	
11	1	0	1	1	1	
12	1	1	0	0	0	
13	1	1	0	1	1	
14	1	1	1	0	00	
15	1	1	1	1	0	

U = 1 AND (L = 0 OR H = 0) AND V = 1

	H'V'	H'V	HV	HV'
U'L'	0	1	3	2
U'L	4	5	7	6
UL	12	13	15	14
UL'	8	9	11	10

$$A = UH'V + UL'V$$

UH'V UL'V

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- For an **online** course, if at home, the student must have both a laptop and headphones.
- Students can also attend an **online** class from university if he is vaccinated, or he has a laptop with headphones

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- For an **online** course, if at home, the student must have both a laptop and headphones.

$$\mathbf{U = 0 \text{ AND } (L = 1 \ \& \ H = 1)}$$

- Students can also attend an **online** class from university if he is vaccinated, or he has a laptop with headphones

$$\mathbf{OR \ U = 1 \text{ AND } (V = 1 \text{ OR } (L = 1 \ \& \ H = 1))}$$

B = 1 (Online = 1) When:

U = 0 AND (L = 1 & H = 1)

OR U = 1 AND (V = 1 OR (L = 1 & H = 1))

	U	L	H	V	A	B
U = 0	0	0	0	0	0	0
	0	0	0	1	0	0
	0	0	1	0	0	0
	0	0	1	1	0	0
	0	1	0	0	0	0
	0	1	0	1	0	0
	0	1	1	0	0	1
U = 1	0	1	1	1	0	1
	1	0	0	0	0	0
	1	0	0	(1)	1	1
	1	0	1	0	0	0
	1	0	1	(1)	1	1
	1	1	0	0	0	0
	1	1	0	(1)	1	1
	1	(1)	(1)	0	0	1
	1	1	1	(1)	0	1

	H'V'	H'V	HV	HV'	
U'L'	0	1	3	2	
U'L			1	1	LH
	4	5	7	6	
UL		1	1	1	
	12	13	15	14	
UL'		1	1		
	8	9	11	10	

B = UV+LH

UV